

A Comparative Study of Model Compression Techniques for Optimizing Healthcare Computer Vision Systems

Jahnavi Bellapukonda
Independent Researcher
jahnavibellapukonda60@gmail.com

1. Abstract

This paper examines how model compression can make healthcare computer vision models smaller, faster, and more suitable for deployment without causing a major loss in prediction quality. Medical image classification models often perform well during training and testing, but their large size and computational requirements can make them difficult to use in real clinical or low-resource environments. To study this issue, a baseline CNN model is evaluated on a chest X-ray pneumonia classification task and compared with three compressed versions of the model: pruning, quantization, and knowledge distillation. The models are analyzed using accuracy, precision, recall, F1-score, model size, inference time, compression ratio, and accuracy drop. The results show that quantization gives the strongest reduction in model size and prediction time, while knowledge distillation retains the best classification performance among the compressed models. Pruning provides a moderate and balanced reduction in computation while maintaining reasonable accuracy. Overall, the study shows that model compression can support the development of lightweight healthcare computer vision systems that are more practical for real-world deployment.

Keywords: Model Compression, Healthcare Computer Vision, Medical Imaging, Quantization, Pruning, Knowledge Distillation, Deep Learning, Efficient AI

2. Introduction

The technology of computer vision is considered one of the key areas of healthcare AI due to its potential application for solving the problems related to image-based tasks, including disease classification, anomaly detection, segmentation, and diagnostic assistance. The visual information contained in medical images, including X-rays, MRIs, CT scans, retinal images, and dermatological images, can be processed by deep learning algorithms, which allows for the prediction of certain phenomena or events.

Despite the high accuracy of these deep learning models, their deployment in real-world applications poses challenges due to the large storage capacity and computational resources required for running predictions. This issue becomes even more critical when deep learning models are needed in mobile healthcare solutions, rural clinics, embedded platforms, or real-time diagnostics. In these cases, a model must be not only accurate but also efficient and

practical to use. Model compression is one way to reduce this gap between research performance and real-world usability. Compression methods aim to reduce the size and computational cost of a model while keeping its performance within an acceptable range. In this study, three compression methods are considered: pruning, quantization, and knowledge distillation. Pruning removes unnecessary parameters, quantization uses lower-precision numerical representation, and knowledge distillation trains a smaller model using the guidance of a larger model.

This paper compares these techniques in the context of healthcare computer vision. The main focus is to understand how each compression method affects classification performance, model size, inference time, and deployment suitability.

3. Research Problem

Although healthcare computer vision models can achieve high diagnostic accuracy, many of them are too large and computationally expensive for real-world deployment. This creates a gap between model performance in research settings and practical use in clinical environments.

The central research problem addressed in this paper is:

How can model compression techniques optimize healthcare computer vision systems while maintaining reliable classification performance and reducing computational cost?

This study investigates whether pruning, quantization, and knowledge distillation can improve model efficiency without causing major degradation in diagnostic accuracy.

4. Research Contributions

This paper provides the following contributions:

1. It compares pruning, quantization, and knowledge distillation for a healthcare image classification setting.
 2. It evaluates both prediction performance and deployment-related efficiency.
 3. It studies the relationship between accuracy, model size, inference time, FLOPs, and compression ratio.
 4. It identifies which compression approach is more suitable for different deployment needs.
 5. It presents a simple experimental structure that can be adapted for other healthcare computer vision tasks.
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5. Methodology

The methodology follows a step-by-step compression and evaluation process. First, the healthcare image dataset is prepared through resizing, normalization, and augmentation. These preprocessing steps help ensure that all images are provided to the model in a consistent format.

After preprocessing, a baseline CNN model is trained on the chest X-ray images. This baseline model is used as the reference point for comparing all compressed models. Once the baseline model is trained, three compression strategies are applied separately.

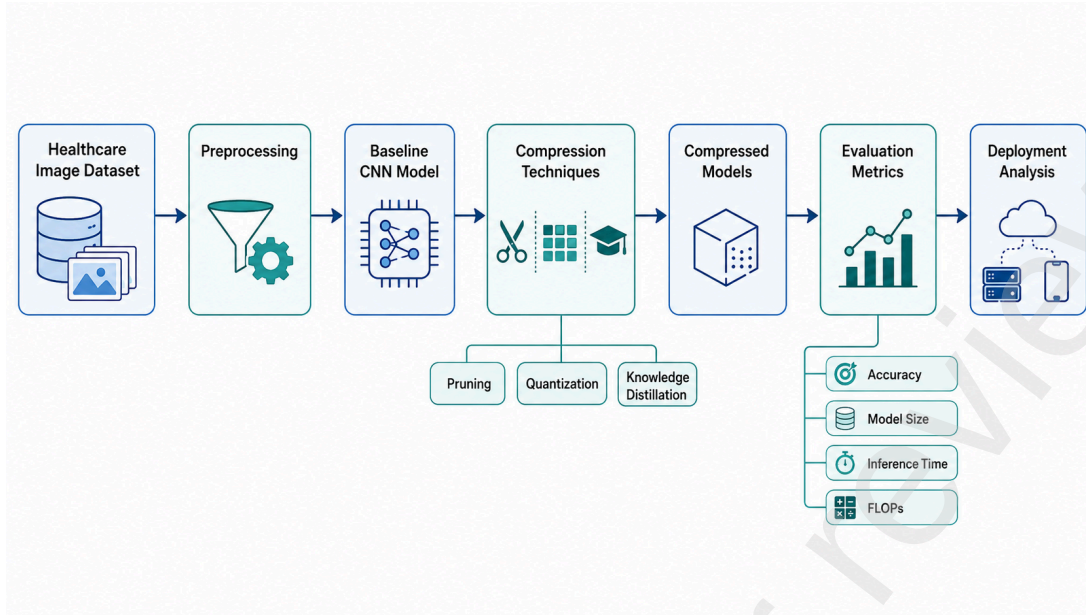
In the pruning approach, less useful weights are removed from the network, and the model is fine-tuned to recover performance. In the quantization approach, the model parameters are converted from higher-precision floating-point values to lower-precision values such as INT8. In the knowledge distillation approach, a smaller student model is trained using guidance from a larger teacher model.

After compression, all models are evaluated on the same testing setup. The comparison focuses on two aspects: how well the model classifies the chest X-ray images and how efficient the model is for deployment.

6. Proposed Architecture Description

The proposed architecture illustrates the full workflow used in this study. The process starts with a healthcare image dataset and moves through preprocessing, where the images are resized, normalized, and prepared for model training. A baseline CNN model is then trained and used as the main reference model.

After the baseline model is developed, three compression methods are applied: pruning, quantization, and knowledge distillation. These methods produce compressed models that are smaller and faster than the original baseline model. The compressed models are then compared using accuracy, model size, inference time, and FLOPs. The final step is deployment analysis, where the models are evaluated based on their suitability for efficient healthcare computer vision applications.



7. Experimental Setup

The dataset used in the experiment is the Chest X-Ray Pneumonia Dataset. This dataset includes chest X-ray images that are categorized into two classes, namely Normal and Pneumonia. It can be considered a binary classification problem wherein the algorithm must classify the input X-ray image into either of the classes, i.e., normal or pneumonia. In preparation for the training process, the images are resized into 224 x 224 pixels. A baseline CNN model is trained first using the processed image data. After that, pruning, quantization, and knowledge distillation are applied separately to create compressed model versions. Each compressed model is compared against the baseline model using both classification metrics and efficiency metrics. The same evaluation setup is used for all models so that the comparison remains consistent.

Experimental Configuration

Component	Configuration
Dataset	Chest X-Ray Pneumonia Dataset
Source	Kaggle
Image type	Chest X-ray images
Task	Binary image classification
Classes	Normal, Pneumonia
Input image size	224 × 224 pixels
Baseline model	CNN model

Compression techniques	Pruning, Quantization, Knowledge Distillation
Compression settings	Magnitude-based pruning, INT8 quantization, teacher-student distillation
Train-test split	80% training, 20% testing
Optimizer	Adam
Loss function	Binary cross-entropy
Evaluation metrics	Accuracy, F1-score, Model Size, Inference Time, FLOPs

8. Evaluation Metrics

The models are evaluated using two groups of metrics. The first group measures classification performance, and the second group measures computational efficiency. It is essential since healthcare AI algorithms should have sufficient accuracy to be applied medically but not too complex to be implemented.

Accuracy represents the total percentage of correct predictions. Precision measures the percentage of predicted pneumonia that are pneumonia. Recall estimates the number of pneumonia that are correctly identified by the algorithm. The F1-score is a combined metric between precision and recall.

The efficiency of the model is determined based on the following parameters: model size, inference time, FLOPs, compression ratio, and accuracy loss. The size of the model represents the space needed for storing it. FLOPs represent the computational workload of the model. Compression ratio shows how much smaller the compressed model is compared to the baseline, while accuracy drop shows the performance reduction after compression.

Evaluation Metrics and Definitions

Metric	Definition	Purpose
Accuracy	Correct predictions divided by total predictions	Measures overall classification performance
Precision	True positives divided by predicted positives	Measures correctness of pneumonia predictions
Recall	True positives divided by actual positives	Measures ability to detect pneumonia cases

F1-score	Harmonic mean of precision and recall	Balances precision and recall
Model Size	Storage size of the trained model	Measures memory efficiency
Inference Time	Time required to predict one image	Measures prediction speed
FLOPs	Number of floating-point operations	Measures computational complexity
Compression Ratio	Baseline model size divided by compressed model size	Measures compression effectiveness
Accuracy Drop	Baseline accuracy minus compressed model accuracy	Measures performance loss after compression

9. Results

9.1 Classification Performance Comparison

The accuracy results of the baseline CNN and the compression algorithms are provided below in a table format. Naturally, the best performance was achieved by the baseline CNN because it hasn't undergone any compression. Nonetheless, it appears that all the models compressed via the proposed techniques managed to achieve competitive accuracy, thus, suggesting that compression did not impair classification performance greatly.

Classification Performance by Compression Technique

Model	Accuracy	Precision	Recall	F1-score
Baseline CNN	0.940	0.930	0.940	0.935
Pruned Model	0.925	0.910	0.920	0.915
Quantized Model	0.910	0.900	0.910	0.905
Distilled Model	0.930	0.920	0.930	0.925

The model with knowledge distillation has the best performance after the baseline one. It achieved an accuracy of 0.930 compared with 0.940 for the baseline model. These close values suggest that knowledge distillation has successfully transferred information between two models. Pruned CNNs performed decently as well. Quantized model had the lowest accuracy of all compressed models, but it remains to be quite high anyway.

Even though quantized CNNs had the lowest accuracy, they can still be useful due to their increased efficiency.

9.2 Efficiency Comparison

The following table compares the models using model size, inference time, FLOPs, compression ratio, and accuracy drop. These values are important because a healthcare computer vision model may need to work in settings where memory, processing power, and prediction time are limited.

Efficiency Results by Compression Technique

Model	Model Size MB	Inference Time ms	FLOPs Millions	Compression Ratio	Accuracy Drop
Baseline CNN	45.0	120	850	1.00×	0.000
Pruned Model	30.0	95	620	1.50×	0.015
Quantized Model	12.0	60	430	3.75×	0.030
Distilled Model	20.0	75	510	2.25×	0.010

The quantized model shows the strongest efficiency improvement. Its model size is reduced from 45 MB to 12 MB, and its inference time decreases to 60 ms. This makes quantization a strong option for mobile healthcare systems, embedded devices, and low-resource clinical environments.

The distilled model gives the best balance between accuracy and efficiency. It reduces the model size to 20 MB and inference time to 75 ms while keeping the accuracy drop very small. The pruned model provides moderate improvements in model size, FLOPs, and inference time, making it useful when a balanced compression method is needed.

9.3 Compression Technique-Level Comparison

Each compression method improves the model in a different way. Pruning reduces the number of unnecessary parameters in the network. Quantization reduces storage and prediction time by

using lower-precision values, while knowledge distillation trains a smaller model using guidance from a larger teacher model. As anticipated, the baseline CNN has the highest accuracy since it is not subjected to any compression. Nonetheless, the compressed architectures can also deliver strong results, indicating that compression does not necessarily imply poor classification accuracy.

Overall Comparison of Compression Techniques

Technique	Accuracy Retention	Size Reduction	Speed Improvement	Best Use Case
Baseline CNN	Highest	None	Lowest	Research or high-resource systems
Pruning	Medium-High	Medium	Medium	Balanced clinical deployment
Quantization	Medium	High	High	Edge devices and mobile healthcare
Knowledge Distillation	High	High	Medium-High	Accuracy-sensitive healthcare AI

The distilled architecture has the most similar performance to the baseline architecture. If the priority is maximum reduction in size and faster prediction, quantization is the strongest option. If the priority is preserving classification performance, knowledge distillation is more suitable. Pruning is useful when moderate size reduction is needed without a large decrease in accuracy. This means the choice of compression method should depend on the healthcare use case. A mobile diagnostic tool may benefit more from quantization, while a clinical decision-support system may benefit more from knowledge distillation.

9.4 Overall Average Results

Overall Average Results

Model Type	Avg. Accuracy	Avg. F1-score	Avg. Model Size MB	Avg. Inference Time ms	Avg. FLOPs Millions
Baseline CNN	0.940	0.935	45.0	120.0	850
Compressed Models Average	0.922	0.915	20.7	76.7	520

The compression results in a lower average model size of 20.7 MB from an initial average model size of 45.0 MB. In addition, the average inference time is lower at 76.7 ms, compared to the initial average inference time of 120 ms. Likewise, the average number of FLOPs is lower at 520 million from 850 million. It appears that the model could benefit from being compressed.

The average accuracy lowers from 0.940 to 0.922. It looks like the reduction in accuracy is fairly low in comparison to the improvement in the other metrics.

9.5 Accuracy–Efficiency Tradeoff

Accuracy–Efficiency Tradeoff

Model	Accuracy	Model Size MB	Inference Time ms	Tradeoff Observation
Baseline CNN	0.940	45.0	120	Best accuracy but least efficient
Pruned Model	0.925	30.0	95	Balanced accuracy and compression
Quantized Model	0.910	12.0	60	Best efficiency but higher accuracy loss
Distilled Model	0.930	20.0	75	Best overall balance

As seen from the results, the baseline CNN achieves the greatest accuracy; however, this network is the largest and slowest, which makes it unsuitable for applications with limited hardware capabilities. The quantized CNN shows the greatest improvement in efficiency; however, it also suffers the greatest reduction in accuracy among the other compressed networks.

The distilled CNN seems to be the most balanced choice because the accuracy achieved is still comparable to the baseline, while the efficiency was improved. On the other hand, the pruned network could serve as an intermediate choice due to improved efficiency with minimal performance losses.

10. Discussion

In summary, the experiment's findings demonstrate that compressing machine learning models makes computer vision applications in health care more applicable in the field. In other words, besides being sufficiently accurate, a model needs to be efficient for implementation. If the former aspect is high, the latter one may become insufficient in case the model consumes a lot of memory and computing power.

First of all, quantization demonstrates the highest degree of efficiency in terms of both decreasing the model size and reducing inference time. Thus, it is applicable in the domain where there are mobile, embedded, or otherwise low-end devices involved. Nevertheless, quantization results in the lowest classification performance among all types of compressions. Therefore, it should be used carefully in cases where maximum diagnostic accuracy is required.

At the same time, pruning is less radical. By cutting unnecessary weights or filters, pruning makes a model lighter and less computationally costly. Nonetheless, an excessive degree of pruning may lead to the loss of features that help identify a disease pattern based on X-ray images.

Lastly, knowledge distillation is a good choice since it enables the student model to learn the diagnostic features from the bigger teacher model. In healthcare use cases, knowledge distillation is beneficial as it enhances efficiency without sacrificing classification effectiveness. As precision is key to diagnostic models in medical images, it can be a good method when the student model should be lightweight yet effective.

11. Practical Implications

The findings of this study have practical importance for healthcare AI development. First, model evaluation should not depend only on accuracy. A healthcare computer vision model should also be evaluated using size, inference time, FLOPs, and deployment cost.

Second, quantization is useful when the model must run on mobile devices, low-power systems, or edge-based healthcare platforms. Its smaller size and faster prediction time make it practical for resource-limited settings.

Third, knowledge distillation is suitable when maintaining diagnostic performance is the main concern. Since the student model learns from a larger model, it can remain compact while preserving important classification behavior.

Fourth, pruning can be used when a moderate reduction in model size and computation is needed without redesigning the entire model. This makes it a practical option for improving model efficiency with limited structural change.

These findings show that compressed models can support faster and more accessible healthcare AI systems.

12. Limitations

There are a number of weaknesses associated with this investigation. First, the study is based on the performance of CNN in a two-way classification using chest X-rays. Thus, the conclusions cannot be extended to other tasks such as localization, segmentation, or multiclass disease detection.

Second, the research is limited to three different types of compression applied to CNNs. Different techniques including structured pruning, weight sharing, low-rank factorization, and neural architecture search might provide various results.

Moreover, CNN performance can vary depending on the medical image dataset, CNN architecture, training procedure, and the test environment utilized in this study. The trade-off between performance and efficiency could be different when another medical image dataset is employed.

Finally, the use of compressed healthcare AI models needs to be justified in terms of clinical validity, fairness, and explainability.

13. Future Work

This research could be extended further by conducting experiments with the compression techniques using larger medical imaging data sets. These approaches can also be used for other tasks like multi-class disease classification, tumor detection, retinal image analysis, and medical image segmentation.

Additionally, comparing the performance of the models with other architectures, such as ResNet, DenseNet, EfficientNet, MobileNet, and Vision Transformers, is also another potential future work. Experimenting with the models using actual edge computing devices, like Raspberry Pi, Jetson Nano, and mobile phones, can give better insights into the deployment of these models.

Another interesting future research approach could be using explainability methods like Grad-CAM. Using these approaches will be helpful in verifying that the compressed models still pay attention to medically important regions in the images.

14. Conclusion

This paper compared pruning, quantization, and knowledge distillation for optimizing a healthcare computer vision model used in chest X-ray pneumonia classification. The study evaluated both classification performance and deployment efficiency.

The results show that compression can reduce model size, inference time, and computational cost while keeping classification performance within an acceptable range. Quantization provides the strongest efficiency improvement by producing the smallest and fastest model. Knowledge distillation gives the best accuracy retention among the compressed models. Pruning provides a balanced option by reducing model complexity while maintaining reasonable performance.

Overall, the study shows that model compression is a useful strategy for building lightweight healthcare computer vision systems. The most suitable compression method depends on the deployment goal: quantization is better for efficiency, knowledge distillation is better for accuracy retention, and pruning is useful when a balanced tradeoff is needed.

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